

WAVEFORM RECONSTRUCTION AS A MEASUREMENT VARIABLE IN PERTURBATION-BASED AUDIO XAI

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ABSTRACT

Perturbation-based evaluation of spectrotemporal explanations may require waveform reconstruction before querying a waveform-input classifier. We isolate this step by comparing original-phase inverse STFT (OP) with 64-iteration Griffin–Lim (GL64), holding the attribution ranking, mask, budget, target magnitude, and downstream query fixed, and analyze two bioacoustic datasets: Curruíra (vocalizations from 49 males) and DB3V (40 source recordings). For retention perturbations in Curruíra, both primary explainers show predominantly negative OP–GL64 contrasts, whereas no primary DB3V comparison remains significant after multiplicity correction. Signal diagnostics confirm distinct waveform realizations of the same target magnitude, with downstream effects shaped by query geometry. A post-primary audit shows that reconstruction can also affect explainer–baseline inference: GEFA’s advantage over an energy-density baseline is supported under OP but not GL64, with a multiplicity-corrected reconstruction interaction, while KernelSHAP changes in magnitude without changing the qualitative conclusion. Waveform reconstruction is therefore part of the faithfulness measurement procedure rather than a transparent conversion step.

Index Terms— audio explainability, faithfulness evaluation, waveform reconstruction, spectro-temporal perturbation, bioacoustics

1. INTRODUCTION

Perturbation-based faithfulness evaluates whether explanation-ranked evidence matters to a model prediction by retaining or removing that evidence and measuring the resulting change in model output [1]. In audio, this evaluation may require an additional reconstruction step when the intervention is defined in a time–frequency (TF) representation but the classifier operates on waveforms. Audio-XAI pipelines therefore commonly reconstruct modified TF representations before the resulting signal is evaluated by the model [4, 5]. However, a modified STFT magnitude does not uniquely determine a waveform, and different phase assumptions or reconstruction procedures can yield different signal realizations [2, 3]. The relevant question for faithfulness evaluation is therefore not whether reconstruction methods can produce different waveforms, but whether these differences propagate to the perturbation-based measure used to evaluate an explanation.

This issue is directly relevant to current audio-XAI pipelines. L-MAC reconstructs magnitude-domain masks using the input phase, whereas LMAC-TD avoids this step by generating explanations directly in the time domain [5, 6]; spectrogram-domain attribution for a waveform-input HuBERT model has likewise been evaluated after

original-phase reconstruction [7]. Removal-based attribution benchmarks show more broadly that evaluation can depend on the intervention protocol rather than ranking alone [16, 8], while iterative phase reconstruction itself depends on the reconstruction procedure and iteration regime [17]. Yet the reconstruction operator has not, to our knowledge, been isolated under an otherwise fixed TF perturbation. We isolate this factor by comparing original-phase inverse STFT (OP) with 64-iteration Griffin–Lim (GL64) while holding the intended intervention fixed. The input, attribution ranking, perturbation direction, budget, binary mask, target TF magnitude, downstream target, and all nonperturbed items are identical across reconstruction methods. We evaluate the comparison in two bioacoustic pipelines with different aggregation schemes: Curruíra [9, 10] and DB3V [11]. We ask whether OP and GL64 realize the same TF intervention differently at the waveform level and, if so, whether those differences propagate to the measured perturbation response.

2. CONTROLLED RECONSTRUCTION

2.1. Fixed spectro-temporal intervention

Let $X = \mathcal{S}(x) = A \odot e^{j\Phi}$ be the STFT of the selected local waveform. Let P denote the interpretable TF domain and $M_b \leq P$ select the highest-ranked $k_b = \max(1, \lceil bK \rceil)$ of its K regions. Rankings use descending signed support for the original predicted class and remain fixed across reconstruction operators. We use *IN* for retention perturbations, where the highest-ranked regions are preserved and the remaining TF regions are attenuated, and *OUT* for removal perturbations, where the highest-ranked regions themselves are attenuated. With attenuation gain α ,

$$\begin{aligned} G_b^{\text{IN}} &= 1 - (1 - \alpha)(P - M_b), \\ G_b^{\text{OUT}} &= 1 - (1 - \alpha)M_b, \quad \tilde{A}_{b,d} = G_b^d \odot A. \end{aligned} \quad (1)$$

Bins outside P retain unit gain in both directions. Thus, for a fixed ranking, direction, and budget, both reconstruction operators receive exactly the same target magnitude $\tilde{A}_{b,d}$.

2.2. Reconstruction and matched faithfulness

We compare two waveform realizations of the same perturbed magnitude:

$$\begin{aligned} x_{\text{OP},b,d} &= \mathcal{S}^{-1}(\tilde{A}_{b,d} e^{j\Phi}), \\ x_{\text{GL64},b,d} &= R_{64}(\tilde{A}_{b,d}; \Phi), \end{aligned} \quad (2)$$

where OP preserves the original phase and R_{64} denotes 64 iterations of Griffin–Lim initialized from Φ . The intended TF intervention is therefore identical across operators; only the waveform reconstruction procedure changes.

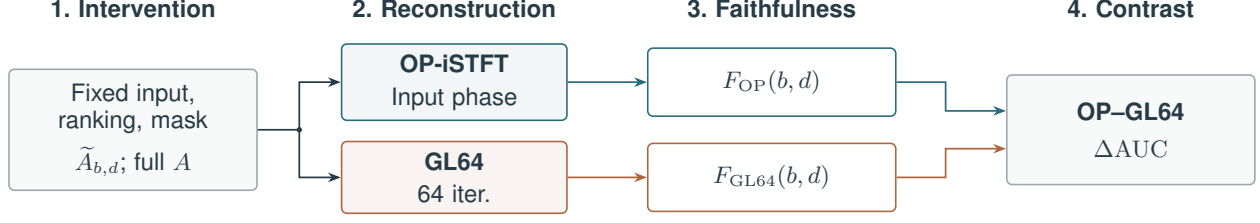


Fig. 1. Controlled reconstruction comparison. Ranking, perturbation direction, mask, budget, and target TF magnitude are fixed before reconstruction; OP and GL64 then realize the same target magnitude before the downstream query and matched faithfulness computation.

To avoid attributing reconstruction-specific reference offsets to the perturbation itself, each operator also reconstructs the corresponding unmasked local item, yielding $x_{R,\text{full}}$. Following the retention/removal logic of sufficiency and comprehensiveness [1], we define the operator-matched faithfulness response as

$$F_R(b, d) = z_t(x_{R,\text{full}}) - z_t(x_{R,b,d}), \quad (3)$$

where $z_t(\cdot)$ is the downstream score for the original target class t .

The reconstruction contrast is summarized across the perturbation-budget set B by

$$\Delta\text{AUC} = \text{trapz}_{b \in B} F_{\text{OP}}(b, d) - \text{trapz}_{b \in B} F_{\text{GL64}}(b, d). \quad (4)$$

Because the downstream score scales differ across pipelines, absolute AUC magnitudes are not compared between datasets. For IN perturbations, $\Delta\text{AUC} < 0$ indicates a smaller operator-matched sufficiency-like gap under OP than under GL64; it does not by itself imply that either reconstruction is intrinsically preferable.

2.3. Downstream query geometry

Waveform-level reconstruction differences need not propagate unchanged to the final faithfulness score because the perturbed local item is embedded within a downstream aggregation rule. For DB3V recording j , the recording representation is the mean of n_j clip embeddings. Replacing one clip therefore changes the pooled representation by $(e_{j\ell,R} - e_{j\ell})/n_j$. Since standardization followed by the target-class logistic-regression decision function is affine,

$$\Delta\text{AUC}_{j,\text{recording}} = \frac{\Delta\text{AUC}_{j,\text{local}}}{n_j}. \quad (5)$$

This identity shows that local reconstruction differences are attenuated exactly by the recording-level averaging factor.

Curruíra follows a different aggregation geometry. For male i with m_i songs, replacing local song ℓ changes the male-level mean probability according to

$$\bar{p}_{i,R}(b, d) = \bar{p}_i + \frac{p_{i\ell,R}(b, d) - p_{i\ell}}{m_i}, \quad z_{i,R} = \text{logit}(\text{clip}(\bar{p}_{i,R})). \quad (6)$$

Away from clipping boundaries, propagation therefore contains both the averaging factor $1/m_i$ and the nonlinear logit sensitivity $1/[\bar{p}_i(1 - \bar{p}_i)]$. These terms describe how reconstruction differences present at the local query level can be transformed before entering the final faithfulness measurement.

3. EXPERIMENTAL SETUP

Dataset Curruíra. We analyze 889 vocalization cuts from 49 male Southern House Wrens, with 24 and 25 males in the higher- and lower-road-noise classes [9]. This corpus was previously studied with XAI for anthropogenic-noise analysis [10]; here it is reused for a controlled reconstruction-measurement question. The frozen query combines BirdNET V2.4 embeddings [12] and acoustic descriptors with class-balanced logistic regression. One deterministic song per male is perturbed while all others remain fixed, and the resulting male-level probability is converted to an oriented logit.

Dataset DB3V. We evaluate cross-region species recognition in DB3V under two transfer directions: D1→D3 (train on D1, test on D3) and D3→D1 (train on D3, test on D1). [11] using 40 correctly classified source recordings from ten species, selected before reconstruction results were available. One deterministic 8-s clip per recording is perturbed. BirdNET clip embeddings are mean-pooled at recording level and classified with class-balanced logistic regression; the original predicted-species decision score defines z_t .

Curruíra uses 44.1-kHz audio and a 7×8 TF proxy, while DB3V uses 16-kHz audio and a 5×8 proxy. Both use a 512-sample Hann STFT, hop 128, centered frames, attenuation $\alpha = 0.01$ (−40 dB), and budgets of 5, 10, 20, 40, 60, and 80%; primary AUCs use 10–80%. GL64 uses 64 deterministic iterations initialized with the original phase and no momentum. KernelSHAP [13] and GEFA [14] are the primary explainers, with rankings fixed across OP and GL64.

Signal diagnostics quantify spectral convergence, preserved-region change, residual attenuated energy, attenuated-region error, and absolute RMS deviation. Males and source recordings are the respective resampling units. Mean contrasts use paired 20,000-replicate bootstrap CIs and 50,000-replicate sign-flip tests [18, 19], with Holm correction over four explainer-by-direction conditions per dataset. Exact sign tests assess directional prevalence, centered studentized-bootstrap tests provide a mean-null sensitivity analysis, and influence is checked by trimming and high-influence deletion.

As a post-primary consequence audit in Curruíra, we compare each explainer with three attribution-free baselines under both reconstruction operators: a *random-ranking* baseline, summarized by the within-male median over 20 random rankings; a *total-energy* baseline, which ranks TF regions by summed spectral energy; and an *energy-density* baseline, which ranks regions by mean spectral energy. Direction-aware advantage is $\text{Adv}_R = \text{AUC}_{c,R} - \text{AUC}_{e,R}$ for IN and $\text{Adv}_R = \text{AUC}_{e,R} - \text{AUC}_{c,R}$ for OUT, so positive values favor the explainer; reconstruction interaction is $\Delta\text{Adv} = \text{Adv}_{\text{OP}} - \text{Adv}_{\text{GL64}}$. Holm correction is applied separately over 24 operator-specific comparisons and 12 interactions; an operator-specific advantage is called resolved only when its bootstrap CI and directional test support the same positive conclusion.

Reproducibility. Executed notebooks, frozen analysis artifacts, and source data supporting the reported tables and figures are available

Table 1. Primary OP–GL64 faithfulness contrasts. Mean CIs and $p_{SF,H}$ are production-analysis values fixed before the post-primary audits; $p_{sign,H}$ tests directional prevalence and $p_{stud,H}$ is a centered studentized-bootstrap mean-null sensitivity. All p values are Holm-adjusted over four explainer-by-direction conditions per dataset.

Dataset	Explainer	Dir.	Mean [95% CI]	Median	Neg./nonzero	$p_{SF,H}$	$p_{sign,H}$	$p_{stud,H}$
Curruíra	GEFA	IN	−0.0556 [−0.1064, −0.0160]	−0.0003	34/48	0.0023	0.0221	0.3322
	GEFA	OUT	0.0721 [−0.0055, 0.1942]	0.0002	22/49	0.1446	0.5682	0.6507
	KernelSHAP	IN	−0.0308 [−0.0607, −0.0081]	−0.0004	34/49	0.0169	0.0282	0.3611
	KernelSHAP	OUT	0.1843 [0.0041, 0.5191]	0.0022	16/49	0.1124	0.0426	0.6507
DB3V	GEFA	IN	−0.0099 [−0.0188, −0.0023]	−0.0048	27/40	0.0797	0.1539	0.1711
	GEFA	OUT	−0.0055 [−0.0136, 0.0028]	−0.0016	24/40	0.4039	0.8046	0.4083
	KernelSHAP	IN	−0.0070 [−0.0153, 0.0005]	−0.0024	24/40	0.2824	0.8046	0.3140
	KernelSHAP	OUT	3.03×10^{-5} [−0.0071, 0.0077]	-5.22×10^{-5}	20/40	0.9940	1.0000	0.9942

at <https://github.com/Ufamalas/xai-wave>.

4. RESULTS

4.1. Faithfulness contrasts and heterogeneity

Curruíra IN has negative mean contrasts for both primary explainers (Table 1). The production percentile-bootstrap CIs exclude zero and the production sign-flip tests survive Holm correction, but the strong mean–median separation motivates the distributional audit. Negative contrasts occur in 34/48 nonzero GEFA cases and 34/49 KernelSHAP cases; exact sign tests remain significant after Holm correction (0.0221 and 0.0282). Centered studentized-bootstrap tests of the mean do not survive Holm (0.3322 and 0.3611). Thus directional prevalence and inference on mean magnitude tell different stories under the observed heavy-tailed heterogeneity.

The negative Curruíra IN mean is not a one-unit artifact. GEFA and KernelSHAP remain negative after 5–10% trimming and after sequential deletion of the five largest $|\Delta\text{AUC}|$ values (top-five means −0.0100 and −0.0099). OUT is substantially more tail-sensitive: KernelSHAP-OUT falls from a mean of 0.1843 to 0.0275 after deleting the largest absolute contrast and to 0.0081 after deleting the five largest; GEFA-OUT falls to 0.00045 after the top-five deletion. KernelSHAP-OUT nevertheless has Holm-corrected positive directional prevalence under the exact sign test while neither mean-oriented test survives correction, underscoring that the tests target different distributional properties. None of the four DB3V explainer-by-direction comparisons remains significant after Holm correction under either the exact sign test or the centered studentized-bootstrap test.

4.2. Consequences for explainer–control comparisons

Reconstruction dependence also propagates to explainer–baseline evaluation in Curruíra. For GEFA-IN versus the energy-density baseline, the mean direction-aware advantage is 0.1791 under OP (95% CI [0.0589, 0.3700]) and 0.1226 under GL64 (−0.0127, 0.3171). Under the conservative decision rule, the advantage is therefore resolved under OP but not under GL64, although GL64 still shows positive directional prevalence. The reconstruction interaction is resolved after multiplicity correction ($\Delta\text{Adv} = 0.0565$, 95% CI [0.0168, 0.1074], $p_H = 0.0154$).

KernelSHAP-IN versus the energy-density baseline remains resolved under both OP and GL64, but its advantage changes in magnitude: $\Delta\text{Adv} = 0.0317$ (95% CI [0.0091, 0.0628], $p_H = 0.0056$). Both resolved interactions remain positive under trimming and high-influence deletion, although their near-zero medians indicate strong unit-level heterogeneity. No random-ranking or OUT reconstruction interaction survives Holm correction. Reconstruction can therefore alter either advantage magnitude or, in one structured compari-

son, the inferential conclusion without producing a general reversal across controls or perturbation directions.

4.3. Signal realization and matched references

OP and GL64 realize the same TF target differently at signal level. Spectral-convergence, residual attenuated-energy, and attenuated-region-error contrasts survive the declared signal-family correction in both datasets; preserved-region differences survive correction only for Curruíra IN. Absolute RMS deviation also differs under its descriptor-family correction, whereas peak frequency, BW90, and frequency-modulation summaries do not. Reduced target-magnitude error under GL64 is consistent with its phase-reconstruction objective and is not interpreted as superior explanation quality interpreted as superior explanation quality (Fig. 2(c)).

Curruíra OP-full and GL64-full oriented scores coincide at stored precision for all 49 males; the original-to-full quality-control drift has maximum absolute value 8.97×10^{-6} . DB3V full-reference offsets are small but nonzero: the median $|z_{OP,full} - z_{GL64,full}|$ is 1.07×10^{-4} , the 95th percentile is 1.53×10^{-3} , and the maximum is 4.45×10^{-3} . Equation (3) therefore compares operator-matched score drops instead of assuming identical full-reference scores.

4.4. Propagation through the downstream query

DB3V follows the affine identity in Eq. (5): a local single-clip contrast is attenuated by $1/n_j$ before the recording-level decision score. Curruíra does not have an analogous affine identity because the mean song probability is followed by clipping and a logit. The aggregation audit nevertheless separates local reconstruction response from downstream geometry. For GEFA IN, $|\Delta\text{AUC}|$ correlates strongly with AUC of the absolute local-song probability gap (Spearman $\rho = 0.835$), compared with $\rho = 0.334$ for $1/m_i$ and $\rho = 0.074$ for local logit sensitivity. KernelSHAP gives $\rho = 0.753$, 0.227, and 0.102, respectively. The largest subject-level contrasts are therefore primarily associated with reconstruction differences already present at the perturbed local-song probability scale, while aggregation modulates their propagation. Supporting RMS matching, iteration-count sweeps, and BELIEF analyses were consistent with the primary qualitative interpretation and are retained in the audit artifact.

5. DISCUSSION AND LIMITATIONS

The controlled comparison establishes that a fixed TF target can yield different measured faithfulness because reconstruction changes the realized waveform before the downstream query. Query geometry conditions this propagation: DB3V imposes exact $1/n_j$ attenuation, whereas Curruíra combines local reconstruction differ-

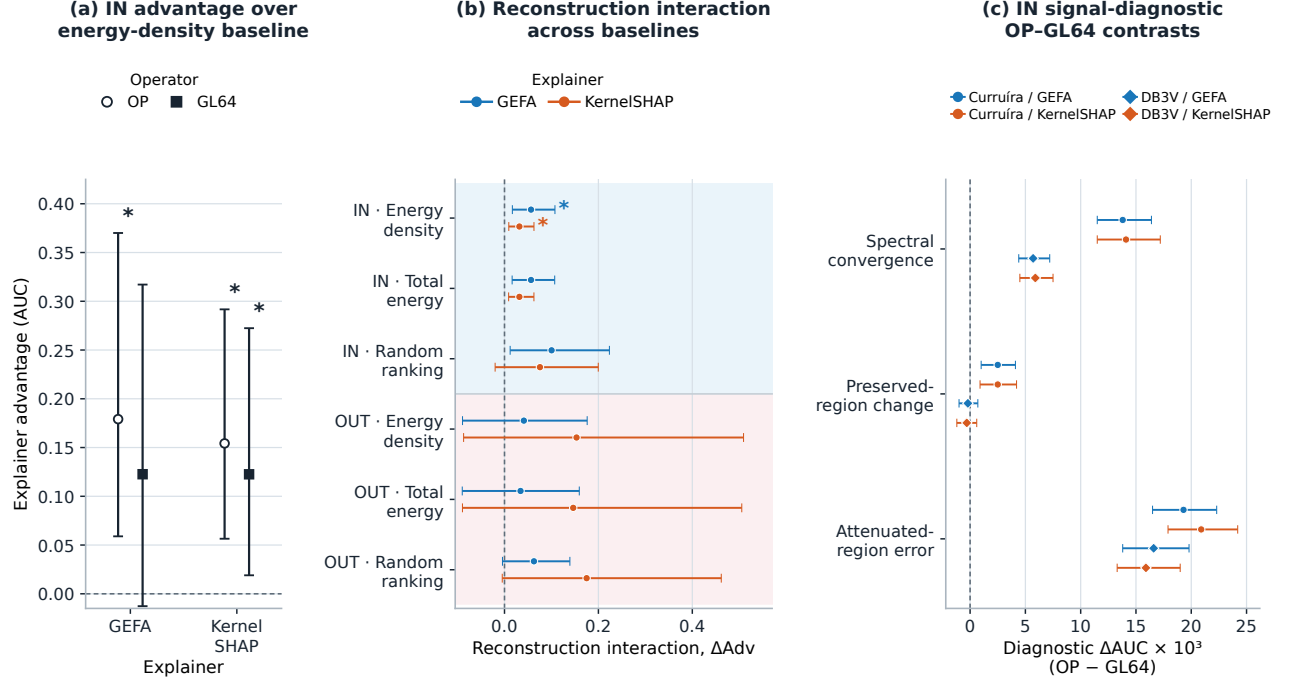


Fig. 2. Reconstruction dependence of perturbation-based audio-XAI evaluation. (a) IN explainer advantage over the energy-density baseline under OP and GL64. (b) Reconstruction interaction across energy-density, total-energy, and random-ranking baselines for GEFA and KernelSHAP. (c) IN OP-GL64 signal-diagnostic contrasts. Error bars show 95% bootstrap CIs; asterisks indicate multiplicity-resolved results within the corresponding inferential family.

ences with averaging and a nonlinear logit. The consequence audit shows that this dependence reaches explainer-baseline inference selectively: the energy-density IN interactions survive Holm correction, while random-baseline and OUT comparisons remain stable. Interpretation therefore remains conditional. Curruíra primary effects are heterogeneous and mean-sensitive, DB3V provides no multiplicity-adjusted primary resolution, and the consequence audit is post-primary and limited to Curruíra. Scope is further restricted to two reconstruction procedures, one attenuation level, two primary explainers, and one local intervention design per dataset.

6. CONCLUSION

Under a matched TF intervention, OP and GL64 receive the same target magnitude but realize different waveforms, and those differences can propagate to perturbation-based faithfulness. Curruíra IN shows a prevalent negative OP-GL64 direction for both primary explainers, whereas no DB3V primary comparison remains resolved after multiplicity correction; the downstream query geometry helps explain why local reconstruction differences propagate differently across the two pipelines. More importantly, the consequence audit shows that reconstruction dependence can reach the conclusion drawn from an explainer-baseline comparison. GEFA’s IN advantage over an energy-density control is resolved under OP but not under GL64, with a Holm-resolved reconstruction interaction. KernelSHAP also shows a significant change in advantage magnitude while retaining the same qualitative conclusion, whereas random-control and OUT comparisons remain stable. Reconstruction dependence is therefore neither negligible nor universal: it varies with the dataset, query geometry, and perturbation type. Perturbation-based audio-XAI evaluation should consequently treat the reconstruction

operator, phase handling, matched references, and downstream aggregation as part of the experimental specification when comparing explanations.

6.1. Use of Generative AI.

OpenAI ChatGPT was used to assist with manuscript revision, LaTeX editing, and visualization-code drafting. The authors remained responsible for the experimental design, scientific decisions, data analysis, interpretation of results, and final manuscript content.

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Compliance with Ethical Standards. This study performs secondary computational analysis of previously published bioacoustic recordings and involved no new animal handling, field collection, or human participants.

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